

Name: _____

Class: _____

YEAR 12
ASSESSMENT TASK 3
TERM 2, 2018

Mathematics Extension 2

General Instructions

- Working time – 90 minutes + 5 minutes reading
- Board approved calculators may be used
- Write using black or blue pen
- All necessary working should be shown in all questions
- One double sided sheet of A4 paper may be brought in and placed on the desk
- Write your student number at the top of every page of your answer sheets.

5 Long Response Questions–12 marks each

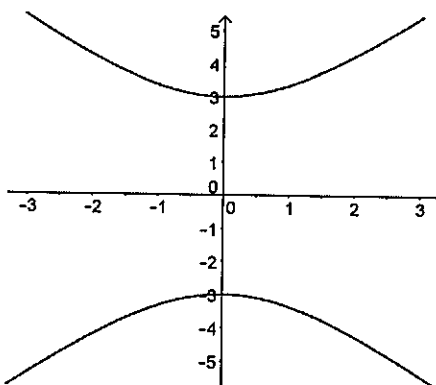
Total marks – 60

Attempt all questions

Start each question on a new sheet of paper.

Question 1 (12 marks)

- a) The region bounded by the lines $x = \pm 2$ and the hyperbola $\frac{y^2}{9} - \frac{x^2}{4} = 1$ is rotated about the y -axis.



- i) Using the method of cylindrical shells, show that $V = 4\pi \int_0^2 x \sqrt{9 + \frac{9}{4}x^2} dx$. 2
- ii) Hence, evaluate the volume of the solid. 2
- b) Points $P(cp, \frac{c}{p})$ and $Q(cq, \frac{c}{q})$ lie on the hyperbola $xy = c^2$.
- i) Show that the equation of the tangent to the hyperbola at P is $x + p^2y = 2cp$. 2
- ii) If the tangents at the points P and Q meet at the point $R(x_0, y_0)$, prove that $pq = \frac{x_0}{y_0}$ and $p + q = \frac{2c}{y_0}$. 3
- iii) If the length of chord PQ is d units, show that $d^2 = c^2(p - q)^2(1 + \frac{1}{p^2q^2})$. 1
- iv) If d remains fixed, deduce that the locus of R has equation $4c^2(x^2 + y^2)(c^2 - xy) = x^2y^2d^2$. 2

Question 2 (12 marks) Start a NEW sheet of paper

- a) A particle is projected vertically upwards in a resisting medium where the resistance varies as the square of the velocity and k is the constant of variation. If the velocity of projection is $v_0 \tan \alpha$,

- i) Show that the equation of motion is given by $\ddot{x} = -kv^2 - g$ where $\ddot{x}$ is the acceleration of the particle and g is the acceleration due to gravity **1**
- ii) Show that the maximum height, H , reached is given by: $H = \frac{1}{2k} \ln\left(\frac{g + kv_0^2 \tan^2 \alpha}{g}\right)$ **3**
- iii) Show that the particle returns to the point of projection with velocity $v_0 \sin \alpha$ given that v_0 is the terminal velocity **4**
- iv) Show that the time of ascent is $\frac{v_0 \alpha}{g}$ **4**

Question 3 (12 marks) Start a NEW sheet of paper

- a) A particle of mass, m kg, moves in a straight line with velocity v metres/second, under a constant force, P newtons, and a resistance R newtons. Initially, the particle has a speed v_0 m/s. It is given that $R = 5 + 3v$ and $P = 10$.

i) Show that $\ddot{x} = \frac{1}{m}(5 - 3v)$. 1

ii) Show that the velocity, $v = \frac{5}{3}(1 - e^{-\frac{3t}{m}}) + v_0 e^{-\frac{3t}{m}}$. 3

iii) Find the terminal velocity of the particle. 1

- iv) When the particle accelerates from v_0 to v_1 , show that the distance travelled,

x metres, is given by: $x = \frac{m}{9}[3(v_0 - v_1) + 5 \ln(\frac{5 - 3v_0}{5 - 3v_1})]$. 3

- b) The area bounded by the lines $y = x + 1$, $x = 2$, $x = 4$ and the x -axis, is rotated about the line $x = 5$. By taking slices perpendicular to the y -axis, find the volume of the solid formed. 4

Question 4 (12 marks) Start a NEW sheet of paper

- a) On a rectangular hyperbola $xy = c^2$, point P has coordinates $(cp, \frac{c}{p})$. It is given

that the equations of the tangent and the normal at this point are

$x + p^2y - 2cp = 0$ and $p^3x - py = c(p^4 - 1)$ respectively. (Do NOT prove these).

- i) Find the coordinates of R where the normal crosses the other branch of the hyperbola. 2

- ii) The coordinates of Q where the tangent meets the y -axis is $(0, \frac{2c}{p})$. Hence show that the area of $\triangle PQR$ is $\frac{c^2}{2p^4}(1 + p^4)^2$. 4

- iii) Hence, find the value(s) of p such that the area is a minimum. 2

- b) Using the letters of the word CALCULUS, how many 'words' can be made if the 'words' consist of:

- i) All eight letters? 1

- ii) Only five letters? 3

Question 5 (12 marks) Start a NEW sheet of paper

- a) The base of a solid is the region bounded by the line $y = 2x$, the parabola

$y = 4x - x^2$ and the x -axis. Cross-sections parallel to the x -axis are right-angled

isosceles triangles with the hypotenuse in the base of the solid.

i) Show that the volume of the solid is given by: $V = \frac{1}{16} \int_0^4 (4 + 2\sqrt{4-y} - y)^2 dy$ 3

ii) Hence, find the exact volume of the solid 2

- b) Two players A and B play a game of chance comprising several turns in which each player throws a fair 6-sided die. The possible outcomes are a draw (A, B throw the same score), A wins (A's score is higher than B's) or B wins. The game is over when either player first records two wins. Let p_n be the probability the game ends on the n^{th} turn. Let q_n be the probability the game does not end in n or fewer turns.

i) Explain why, on each turn, the probability that A wins is $\frac{5}{12}$ 1

ii) Explain why $p_2 + q_2 = 1$ 1

iii) Explain why $p_n + q_n = q_{n-1}$, for $n = 3, 4, 5, \dots$ and deduce

that $\sum_{k=2}^n p_k = 1 - q_n$, $n \geq 2$ 2

iv) Show that $q_n = \frac{25n^2 - 5n + 4}{4 \times 6^n}$ and $p_n = \frac{25(n-1)(5n-8)}{4 \times 6^n}$, $n \geq 2$ 3

End of examination

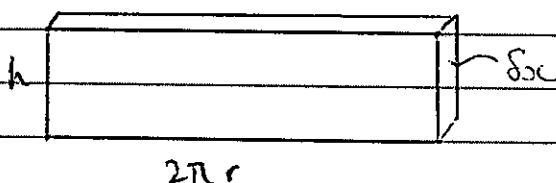
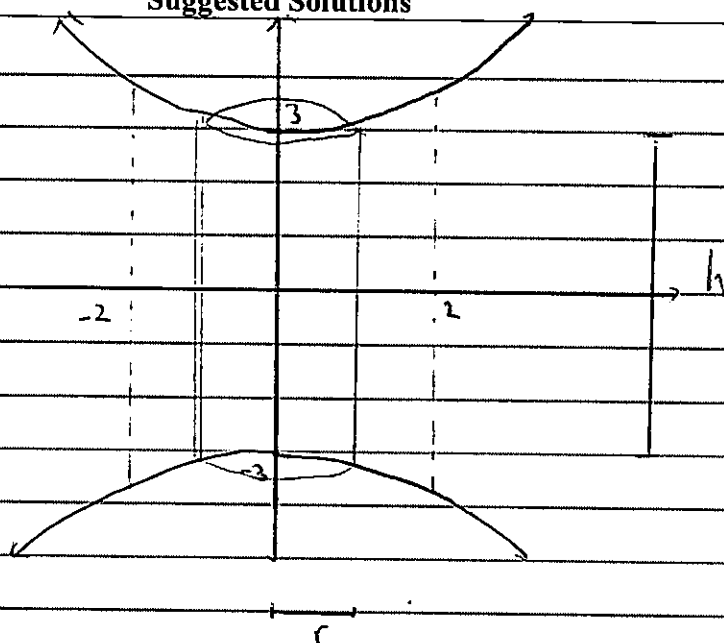
MATHEMATICS Extension 2: Question 1...

Suggested Solutions

Marks

Marker's Comments

a) i)



$$h = 2|y| \quad r = |x|$$

$$\frac{y^2}{9} - \frac{x^2}{4} = 1$$

$$\begin{aligned} \therefore V_{\text{shell}} &= 2\pi r h \delta x \\ &= 2\pi |x| \cdot 2|y| \delta x \\ &= 4\pi |x| \sqrt{9 + \frac{9x^2}{4}} \delta x \end{aligned}$$

$$\frac{y^2}{9} = 1 + \frac{x^2}{4}$$

$$\begin{aligned} y^2 &= 9\left(1 + \frac{x^2}{4}\right) \\ &= 9 + \frac{9x^2}{4} \end{aligned}$$

Taking limits from $x=2$ to $x=0$.

$$\therefore |y| = \sqrt{9 + \frac{9x^2}{4}}$$

$$\begin{aligned} \therefore V_{\text{solid}} &= \lim_{\delta x \rightarrow 0} \sum_0^2 4\pi x \sqrt{9 + \frac{9x^2}{4}} \delta x \\ &= 4\pi \int_0^2 x \sqrt{9 + \frac{9x^2}{4}} dx \end{aligned}$$

Must have limits!!!

①

①

MATHEMATICS Extension 1 : Question.....

Suggested Solutions	Marks	Marker's Comments
ii) $V = 4\pi \int_0^2 x \left(9 + \frac{9}{4}x^2\right)^{\frac{1}{2}} dx$ $= \frac{8\pi}{9} \int_0^2 \frac{9x}{2} \left(9 + \frac{9}{4}x^2\right)^{\frac{1}{2}} dx$ $= \frac{8\pi}{9} \left[\frac{2 \left(9 + \frac{9}{4}x^2\right)^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^2$ $= \frac{16\pi}{27} \left[18^{\frac{3}{2}} - 9^{\frac{3}{2}} \right]$ $= \frac{16\pi}{27} [18\sqrt{18} - 9\sqrt{9}]$ $= \frac{16\pi}{27} (54\sqrt{2} - 27)$ $= 16\pi (2\sqrt{2} - 1)$	① ①	
b) i) $y = \frac{c^2}{x}$ $\frac{dy}{dx} = -c^2 x^{-2}$ When $x = cp$, $\frac{dy}{dx} = \frac{-c^2}{c^2 p^2} = \frac{-1}{p^2}$ $\therefore$ tangent at P is given by: $y - \frac{c}{p} = \frac{-1}{p^2} (x - cp)$ $py - cp = -x + cp$ $\therefore x + p^2 y = 2cp$	① ①	

MATHEMATICS Extension 1 : Question.....

Suggested Solutions

Marks

Marker's Comments

ii) Similarly, tangent at $Q(cq, \frac{c}{q})$ is given by

$$x + q^2 y = 2cq \text{ - Eqn 2}$$

Eqn 1 - Eqn 2



$$\begin{aligned} x + p^2 y &= 2cp \\ - x + q^2 y &= 2cq \\ \hline y(p^2 - q^2) &= 2c(p - q) \end{aligned}$$

$$\therefore y(p + q) = 2c$$

$$y = \frac{2c}{p+q} \quad \therefore y_0 = \frac{2c}{p+q}$$

Sub $y_0 = \frac{2c}{p+q}$ into Eqn 1.

$$x_0 + p^2 \left(\frac{2c}{p+q} \right) = 2cp$$

$$(p+q)x_0 + 2cp^2 = 2cp(p+q)$$

$$\therefore px_0 + qx_0 + 2cp^2 = 2cp^2 + 2cpq$$

$$\therefore x_0(p+q) = 2cpq$$

$$\therefore pq = \frac{x_0(p+q)}{2c}$$

$$= \frac{x_0}{2c/p+q}$$

$$= \frac{x_0}{y_0}$$

and

$$p+q = \frac{2c}{y_0}$$

(1)

(1)

(1)

MATHEMATICS Extension 2: Question.....

Suggested Solutions	Marks	Marker's Comments
$\text{iii) } d^2 = (cp - cq)^2 + \left(\frac{c}{p} - \frac{c}{q}\right)^2$ $= c^2(p-q)^2 + c^2\left(\frac{1}{p} - \frac{1}{q}\right)^2$ $= c^2(p-q)^2 + c^2\left(\frac{q-p}{pq}\right)^2$ $= c^2(p-q)^2 + c^2\left(\frac{p-q}{pq}\right)^2 \quad (\text{Since } (p-q)^2 = (q-p)^2)$ $= c^2(p-q)^2 \left(1 + \frac{1}{p^2q^2}\right)$		must see this line!!!
$\text{iv) } x_0 = \frac{2cpq}{p+q} \quad y_0 = \frac{2c}{p+q}$		
$(p-q)^2 = (p+q)^2 - 4pq$ $= \frac{4c^2}{y_0^2} - \frac{4x_0}{y_0}$ $\frac{1}{p^2q^2} - \frac{1}{(pq)^2} = \frac{y_0^2}{x_0^2}$		①
$\therefore d^2 = c^2 \left(\frac{4c^2}{y_0^2} - \frac{4x_0}{y_0} \right) \left(1 + \frac{y_0^2}{x_0^2} \right)$ $= c^2 \left(\frac{4c^2 - 4x_0 y_0}{y_0^2} \right) \left(\frac{x_0^2 + y_0^2}{x_0^2} \right)$ $\therefore x_0^2 y_0^2 d^2 = c^2 (4c^2 - 4x_0 y_0) (x_0^2 + y_0^2)$ $\therefore 4c^2 (x_0^2 + y_0^2) (c^2 - x_0 y_0) = x_0^2 y_0^2 d^2$		
$\therefore R \text{ has locus } 4c^2 (x^2 + y^2) (c^2 - xy) = x^2 y^2 d^2$		①

X2 MATHEMATICS: Question 2

Suggested Solutions

Marks

Marker's Comments

↑ +ve ↓ kv^2 ↓ mg

(i) $m\ddot{x} = -kv^2 - mg$ since $m=1$
 $\ddot{x} = -kv^2 - g$

1

Students failed to indicate positive direction or did not have $m\ddot{x} =$ expression.

(ii) $\ddot{x} = v \frac{dv}{dx}$

$v \frac{dv}{dx} = -kv^2 - g$

$\int_{v_0 \text{ and } 0}^H \frac{v dv}{kv^2 + g} = \int_0^H -dx$

1

including correct limits

$\therefore -H = \frac{1}{2k} \left[\ln(kv^2 + g) \right]_{v_0 \text{ and } 0}^0$

1

Some students had issue with negative or limit around wrong way.

$\therefore H = \frac{1}{2k} \left[\ln(kv^2 + g) \right]_0^{v_0 \text{ and } 0}$

$= \frac{1}{2k} \left[-\ln g + \ln(kv_0^2 + g) \right]$

$kv^2 + g > 0 \quad = \frac{1}{2k} \left[\frac{\ln(kv_0^2 + g)}{g} \right]$

1

A show question this step must appear.

could also use $+C$

X2 MATHEMATICS: Question 2

Suggested Solutions

Marks

Marker's Comments

$\uparrow kv^2$
 $\downarrow mg$ $\downarrow +ve$ OR $\uparrow kv^2$ $\uparrow +ve$
 $\downarrow mg$

$$m\ddot{x} = mg - kv^2 \quad m=1$$

$$\ddot{x} = g - kv^2$$

$$\int_0^v \frac{v \, dv}{g - kv^2} = \int_0^H dx$$

$\ddot{x} = kv^2 - g$
 but limits
 will be
 different

$$H = -\frac{1}{2k} \left[\ln(g - kv^2) \right]_0^v \quad g - kv^2 > 0$$

$$= -\frac{1}{2k} \left[\ln(g - kv^2) - \ln(g) \right]$$

$$= \ln\left(\frac{g}{g - kv^2}\right)$$

v_0 is terminal velocity

$$\ddot{x} = 0 \quad g - kv_0^2 = 0$$

$$\therefore v_0^2 = \frac{g}{k}$$

equate H from part (i)

$$\frac{1}{2k} \ln\left(\frac{g}{g - kv^2}\right) = \frac{1}{2k} \ln\left(\frac{g + kv_0^2 t_0^2}{g}\right)$$

$$\therefore \frac{g}{g - kv^2} = \frac{g + kv_0^2 t_0^2}{g}$$

Student didn't find positive direction.

MATHEMATICS: Question

Suggested Solutions

Marks

Marker's Comments

$$g - kv^2 = \frac{g^2}{g + kv_0^2 \tan^2 \alpha}$$

$$\text{but } v_0^2 = \frac{g}{k}$$

$$\therefore g - kv^2 = g - \frac{g^2}{g + g \tan^2 \alpha}$$

$$kv^2 = \frac{g^2 + g^2 \tan^2 \alpha - g^2}{g + g \tan^2 \alpha}$$

$$v^2 = \frac{g(1 + \tan^2 \alpha)}{k(1 + \tan^2 \alpha)}$$

$$\text{Now } 1 + \tan^2 \alpha = \sec^2 \alpha$$

$$= \frac{g}{k} \times \cos^2 \alpha \times \frac{\sin^2 \alpha}{\cos^2 \alpha}$$

$$v = v_0 \sin^2 \alpha$$

$$\therefore v = v_0 \sin \alpha \text{ as } v \rightarrow 0$$

1

Student failed
to read question
carefully and
~~failed~~ did not
use this fact

careless algebraic
errors losing
g's and k's.

1

MATHEMATICS: Question

Suggested Solutions

Marks

Marker's Comments

$$\ddot{x} = -kv^2 - g$$

$$\frac{dv}{dt} = -(kv^2 + g)$$

$$-\int_{v_0 \text{ and}}^0 \frac{1}{kv^2 + g} = \int_0^T t \, dt$$

$$-T = \frac{1}{k} \left[\frac{\sqrt{k}}{\sqrt{g}} \tan^{-1} \frac{v}{\sqrt{g/k}} \right]_{v_0 \text{ and}}^0$$

$$= \frac{1}{\sqrt{k}g} \tan^{-1} \frac{\sqrt{k}}{\sqrt{g}} v_0 \tan \alpha - \tan^{-1} 0$$

$$\text{but } v_0 = \sqrt{\frac{g}{k}}$$

$$= \frac{1}{\sqrt{k}g} \tan^{-1} \frac{\sqrt{k}}{\sqrt{g}} \times \frac{\sqrt{g}}{k} \tan \alpha$$

$$= \frac{1}{\sqrt{k}g} \tan^{-1}(\tan \alpha)$$

$$= \frac{\sqrt{g}}{\sqrt{k}} \times \frac{1}{g} \alpha$$

$$= \frac{v_0}{g} \alpha$$

NB

$$\tan^{-1}(2 \tan \alpha)$$

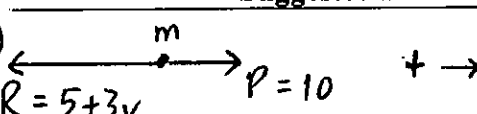
$$\neq 2$$

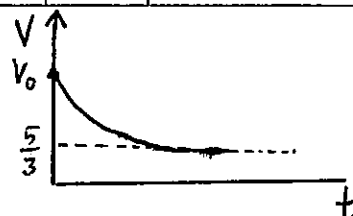
$\therefore$ had to clearly indicate

$$\tan^{-1} \frac{\sqrt{k} v_0 \tan \alpha}{g}$$

$$= \tan^{-1}(\tan \alpha)$$

$$\text{not } \frac{\sqrt{k} v_0}{g} \alpha$$

Task 3: Term 2: 2018 MATHEMATICS Extension 2: Question 3		JRAHS
Suggested Solutions	Marks	Marker's Comments
a) (i)  $P=10$ $\rightarrow$ $m\ddot{x} = F_{\text{net}} \dots$ (Newton's 2nd law) $= P - R$ $= 10 - 5 - 3v$ $\therefore \ddot{x} = \frac{1}{m}(5 - 3v)$	1	For correct derivation Note: when $v=0$, $R=5 < P$ $v=1$, $R=8 < P$ $v=2$, $R=11 > P$ $v=\frac{5}{3}$, $R=10=P$ $v=v_0$, $R=5+v_0$? P
(ii) $\frac{dv}{dt} = \frac{1}{m}(5 - 3v)$ $\int_0^t dt = \frac{1}{m} \int_{v_0}^v \frac{dv}{5-3v}$ $\Rightarrow t = -\frac{m}{3} \ln 5-3v \Big _{v_0}^v$ $\Rightarrow \ln(5-3v) - \ln(5-3v_0) = -\frac{3t}{m}$ $\Rightarrow \ln\left(\frac{5-3v}{5-3v_0}\right) = -\frac{3t}{m}$ i.e. $\frac{5-3v}{5-3v_0} = e^{-3t/m}$ $5-3v = 5e^{-3t/m} + 3v_0e^{-3t/m}$ $3v = 5(1-e^{-3t/m}) + 3v_0e^{-3t/m}$ $\therefore v = \frac{5}{3}(1-e^{-3t/m}) + v_0e^{-3t/m}$ as reqd. $\rightarrow$	1 1 1	For splitting differential Eq. and integrating For integrating Algebraic manipulation Note: If $5-3v < 0$, then same v is obtained!!!



Task 3 Term 2: 2018 MATHEMATICS Extension 2: Question 3			TRAHS
Suggested Solutions	Marks	Marker's Comments	
a) (iii) Terminal velocity reached when $\ddot{x} = 0 \Rightarrow v_T = \frac{5}{3} \text{ m.s}^{-1}$	1	Note $v \rightarrow v_T$ when $t \rightarrow \infty$ $\Rightarrow e^{-3t/m} \rightarrow 0$	
(iv) $v \frac{dv}{dx} = \frac{1}{m} (5 - 3v)$ $\Rightarrow \int_0^x dx = \int_{v_0}^{v_1} \frac{v m dv}{5 - 3v}$ $\Rightarrow \frac{x}{m} = -\frac{1}{3} \int \frac{5 - 3v - 5}{5 - 3v} dv$ $\therefore x = -\frac{m}{3} \int_{v_0}^{v_1} \left[1 - \frac{5}{5 - 3v} \right] dv$ $= -\frac{m}{3} \left[v + \frac{5}{3} \ln(5 - 3v) \right] \Big _{v_0}^{v_1}$ $= -\frac{m}{9} \left[3(v_1 - v_0) + 5 \ln \left(\frac{5 - 3v_1}{5 - 3v_0} \right) \right]$ $= \frac{m}{9} \left[3(v_0 - v_1) + 5 \ln \left(\frac{5 - 3v_0}{5 - 3v_1} \right) \right]$ as reqd.	1 1 1	Note: $\frac{v}{5 - 3v} = A + B$ $\Rightarrow v = A(5 - 3v) + B(5 - 3v)$ $\Rightarrow A = -\frac{1}{3} \text{ and } B = +\frac{5}{3}$ Splitting integral For actual integration For algebraic manipulation	

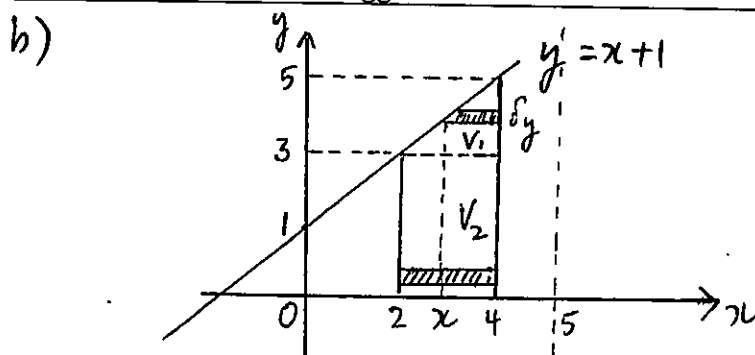
Task 3 : Term 2 : 2018 MATHEMATICS Extension 2: Question 3

JRAHS

Suggested Solutions

Marks

Marker's Comments



Consider region V_1 :

$$\delta V_1 = \pi [(5-x)^2 - 1^2] \delta y$$

$$= \pi [(6-y)^2 - 1] \delta y$$

$$\therefore V_1 = \lim_{\delta y \rightarrow 0} \sum_{y=3}^5 \pi [(6-y)^2 - 1] \delta y$$

$$= \pi \int_3^5 (35 - 12y + y^2) dy$$

$$= \pi (35y - 6y^2 + y^3/3) \Big|_3^5$$

$$= \frac{20\pi}{3} u^3$$

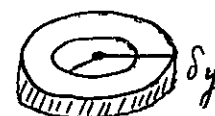
Consider region V_2 :

$$V_2 = \int_0^3 \pi [(5-2)^2 - (5-4)^2] dy$$

$$= 24\pi u^3$$

$$\therefore \text{Total Volume} = \frac{20\pi}{3} + 24\pi$$

$$= 30\frac{2}{3} \pi u^3$$



$$r_{\text{outer}} = 5 - x = 6 - y$$

$$r_{\text{inner}} = 5 - 4 = 1$$

For correctly setting up Volume of slice $\perp$ axis (zero if $\parallel$ to axis)

For correct computation

For volume of cylinder with hole

For combined volume

Note: if slice $\parallel$ axis of rotation
max 2 of 4
if $V_T = \frac{92}{3} \pi u^3$

MATHEMATICS Extension 2: Question...4...

Suggested Solutions

Marks

Marker's Comments

a) i) normal: $p^3x - py = c(p^4 - 1)$ - (1)

$$xy = c^2$$

$$y = \frac{c^2}{x} \quad - (2)$$

sub. (2) into (1)

$$p^3x - p\left(\frac{c^2}{x}\right) = c(p^4 - 1)$$

$$p^3x^2 - pc^2 = xc(p^4 - 1)$$

$$p^3x^2 - c(p^4 - 1)x - pc^2 = 0$$

sum of roots: (let roots be x_1, x_2)

$$x_1 + x_2 = \frac{c(p^4 - 1)}{p^3}$$

one root is cp .

$$\therefore x_1 + cp = \frac{cp^4 - c}{p^3}$$

$$x_1 + cp = cp - \frac{c}{p^3}$$

$$\therefore x_1 = -\frac{c}{p^3}$$

$$\therefore y = \frac{c^2}{-\frac{c}{p^3}}$$

$$= -cp^3$$

$$\therefore R \text{ is } \left(-\frac{c}{p^3}, -cp^3\right)$$

ii) $P\left(cp, \frac{c}{p}\right) \quad Q\left(0, \frac{2c}{p}\right)$

$$d_{PQ} = \sqrt{\left(cp\right)^2 + \left(\frac{c}{p} - \frac{2c}{p}\right)^2}$$

$$= \sqrt{c^2p^2 + \frac{c^2}{p^2}}$$

$$= \sqrt{c^2\left(p^2 + \frac{1}{p^2}\right)}$$

$$= c \sqrt{\frac{p^4 + 1}{p^2}}$$

$$= \frac{c}{p} \sqrt{p^4 + 1}$$

1

1

1

must show that c is outside of square root.

MATHEMATICS Extension 2: Question....4...

Suggested Solutions

Marks

Marker's Comments

$$d_{PR} = \sqrt{\left(cp + \frac{c}{p^3}\right)^2 + \left(\frac{c}{p} + cp^3\right)^2}$$

$$= \sqrt{c^2 \left(\frac{p^4+1}{p^3}\right)^2 + c^2 \left(\frac{1+p^4}{p}\right)^2}$$

$$= c \sqrt{\frac{(p^4+1)^2}{p^6} + \frac{(p^4+1)^2}{p^2}}$$

$$= c(p^4+1) \sqrt{\frac{1}{p^6} + \frac{1}{p^2}}$$

$$= c(p^4+1) \sqrt{\frac{1}{p^2} \left(\frac{1}{p^4} + 1\right)}$$

$$= \frac{c(p^4+1)}{p} \sqrt{\frac{1+p^4}{p^4}}$$

$$= \frac{c(p^4+1)}{p^3} \sqrt{p^4+1}$$

$$\therefore \text{Area} = \frac{1}{2} \times d_{PR} \times d_{PR}$$

$$= \frac{1}{2} \times \frac{c}{p} \sqrt{p^4+1} \times \frac{c(p^4+1)}{p^3} \sqrt{p^4+1}$$

$$= \frac{c^2}{2p^4} (p^4+1)(p^4+1)$$

$$= \frac{c^2}{2p^4} (p^4+1)^2$$

NOTE: Perpendicular distance:

$$x + p^2y - 2cp = 0 \quad R\left(-\frac{c}{p^3}, cp^3\right)$$

$$d = \frac{\left| -\frac{c}{p^3} + p^2(-cp^3) - 2cp \right|}{\sqrt{1^2 + p^4}}$$

$$= \frac{\left| -\frac{c}{p^3} - cp^5 - 2cp \right|}{\sqrt{1+p^4}}$$

$$= \frac{\left| -c(1+p^8+2p^4) \right|}{p^3 \sqrt{1+p^4}}$$

some factorising or expanding needed

correct formula and substitution.

maximum 3 marks.

Could not explain why a negative sign was needed, since p can be positive or negative.

MATHEMATICS Extension 2: Question....4...

Suggested Solutions

Marks

Marker's Comments

$$\text{iii) } A = \frac{c^2}{2p^4} (1+p^4)^2$$

$$\frac{dA}{dp} = \frac{-4c^2}{2p^5} (1+p^4)^2$$

$$+ \frac{c^2}{2p^4} \times 2(1+p^4) \times (4p^3)$$

$$= \frac{-2c^2}{p^5} (1+p^4)^2 + \frac{4c^2}{p} (1+p^4)$$

$$= \frac{2c^2}{p} (1+p^4) \left(-\frac{(1+p^4)}{p^4} + 2 \right)$$

for stationary points, $\frac{dA}{dp} = 0$.

$$\therefore \frac{2c^2}{p} (1+p^4) \left(\frac{-1-p^4+2p^4}{p^4} \right) = 0$$

$$\therefore 1+p^4=0, \quad \frac{p^4-1}{p^4}=0$$

no solution $\therefore p^4-1=0$

$$p^4=1$$

$$p = \pm 1$$

$$\frac{dA}{dp} = \frac{2c^2}{p^5} (1+p^4) (p^4-1)$$

$$= \frac{2c^2}{p^5} (p^8-1)$$

$$\therefore \frac{d^2A}{dp^2} = \frac{-10c^2}{p^6} (p^8-1) + \frac{2c^2}{p^5} (8p^7)$$

$$= -10c^2 p^2 + \frac{10c^2}{p^6} + 16c^2 p^2$$

$$= 6c^2 p^2 + \frac{10c^2}{p^6}$$

$$\text{when } p=1, \quad \frac{d^2A}{dp^2} = 6c^2 + 10c^2 = 16c^2 > 0$$

$\therefore p=1$ is a minimum as it concaves up.

1

MATHEMATICS Extension 2: Question...4...

Suggested Solutions

Marks

Marker's Comments

$$\text{when } p = -1, \frac{d^2A}{dp^2} = 6c^2 + 10c^2 = 16c^2 > 0$$

$\therefore$ concave up

$\therefore p = -1$ is a minimum

minimum at $p = \pm 1$

b) i) $2C_2 \cdot 2L \cdot 2U$ AS

$$\frac{8!}{2!2!2!} = 5040$$

ii) 1: 5 different letters: $5! = 120$

2: one double, 3 different

$$\frac{5!}{2!} \times {}^3C_1 \times {}^4C_3 = 720$$

3 doubles 4 remaining

to choose from letters to choose

3: two doubles, 1 different

$$\frac{5!}{2!2!} \times {}^3C_2 \times {}^3C_1 = 270$$

3 double, 3 letters remaining,

choose 2 choose 1.

$$\therefore \text{total} = 120 + 720 + 270 = 1110$$

1 $\Rightarrow$ testing $p = \pm 1$

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1 $\Rightarrow$ one case correct

1 $\Rightarrow$ another case correct

1 $\Rightarrow$ third case correct and total correct.

MATHEMATICS Extension 2: Question.....5

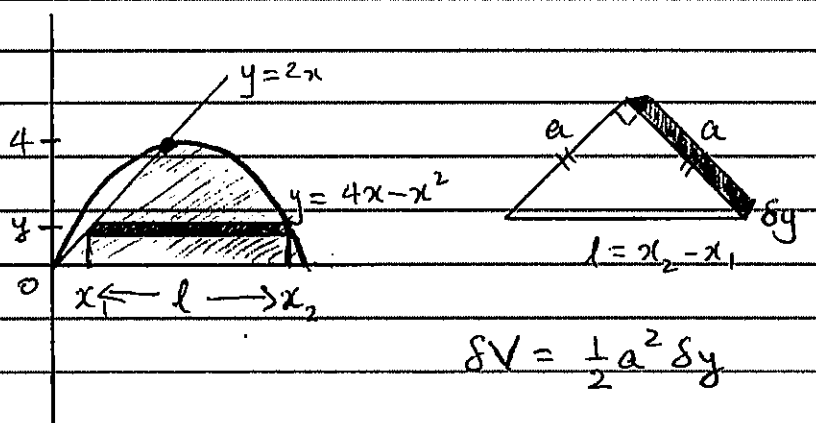
Suggested Solutions

Marks

Marker's Comments

(a)

(i)



$$\delta V = \frac{1}{2} a^2 \delta y$$

$$\text{i.e. } \delta V = \frac{l^2}{4} \delta y$$

since $2a^2 = l^2$ by Pythag. theorem.

$$\text{At } y = 4x - x^2 \Rightarrow x = 2 \pm \sqrt{4 - y}$$

$$\text{Hence } x_2 = 2 + \sqrt{4 - y} \text{ (want } x_2 > x_1)$$

Also

$$y = 2x_1 \Rightarrow x_1 = y/2$$

$$\therefore l = x_2 - x_1$$

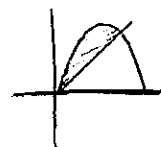
$$= 2 + \sqrt{4 - y} - y/2$$

$$= \frac{1}{2} (4 + 2\sqrt{4 - y} - y)$$

So

$$\delta V = \frac{l^2}{4} = \frac{1}{16} (4 + 2\sqrt{4 - y} - y)^2 \delta y$$

Many identified incorrect region



and then attempted to 'fit' their integrand to that required.

1 1

MATHEMATICS Extension 2: Question...5....

Suggested Solutions

Marks

Marker's Comments

Now, limits: $y = 0$ to y at point of intersection:

$$2x = x(4-x)$$

$$\Rightarrow x(x-2) = 0$$

$$\text{So } x = 0, x = 2$$

$$\text{So } y = 0, y = 4.$$

$\therefore$ limits: $y = 0$ to 4 .

$$\text{So } V = \lim_{\delta y \rightarrow 0} \sum_{y=0}^4 \frac{1}{16} (4 + 2\sqrt{4-y} - y)^2 \delta y$$

$$= \frac{1}{16} \int_0^4 (4 + 2\sqrt{4-y} - y)^2 dy$$

$$(ii) V = \frac{1}{16} \int_0^4 ((4-y) + 2(4-y)^{1/2})^2 dy$$

$$= \frac{1}{16} \int_0^4 (4-y)^2 + 4(4-y)^{3/2} + 4(4-y) dy$$

$$= \frac{1}{16} \left[-\frac{(4-y)^3}{3} - 4 \cdot \frac{2}{5} (4-y)^{5/2} + \frac{4(4-y)^2}{2} \right]_0^4$$

$$= \frac{1}{16} \left[\frac{4^3}{3} + \frac{8}{5} \cdot 4^{5/2} + 2 \cdot 4^2 \right]$$

$$= \frac{98}{15} \text{ units}^3.$$

Limits were not calculated explicitly by many candidates.

You must establish all components of the integral.

Majority of candidature did not obtain correct integral or had trouble integrating their expansion.

MATHEMATICS Extension 2: Question.....5

Suggested Solutions

Marks

Marker's Comments

(b) (i) Let $D \equiv$ draw

$A \equiv$ A wins turn

$B \equiv$ B wins turn

In one turn, either

A or B or D

all mutually exclusive. So,

$$1 = P(A \text{ or } B \text{ or } D)$$

$$= P(A) + P(B) + P(D)$$

$$= 2P(A) + \frac{1}{6} \quad \left(\begin{array}{l} P(D) = \frac{1}{6} \\ P(A) = P(B), \\ \text{Symmetry} \end{array} \right)$$

$$\text{so } P(A) = \frac{5}{12} //$$

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(ii) $q_2 = P(\text{game does not end by 2nd turn})$

$= P(\text{game does not end at 1st turn}$
&

game does not end at 2nd turn)

$= 1 - P(\text{game ends on 1st turn or}$
game ends on 2nd turn)

$= 1 - [P(\text{game ends on 1st turn})$

$+ P(\text{game ends on 2nd turn})$

$- P(\text{game ends on 1st turn \&$

game ends on 2nd turn})

$$= 1 - [0 + p_2 - 0]$$

$$= 1 - p_2 \longrightarrow p_2 + q_2 = 1.$$

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Diagrammatic reasoning may be used - also accepted.

MATHEMATICS Extension 2: Question.....

Suggested Solutions

Marks

Marker's Comments

(iii) Let $G(n)$ be the event that the game ends on the n^{th} turn, and $E(n)$ be the event that the game does not end by the n^{th} turn.

Now, for $n \geq 2$:

The game ends on the n^{th} turn or does not end by the n^{th} turn

if and only if

the game does not end by the $(n-1)^{\text{th}}$ turn;

$$\text{i.e. } G(n) \cup E(n) = E(n-1)$$

So

$$P(G(n) \cup E(n)) = P(E(n-1))$$

$$\text{i.e. } P(G(n)) + P(E(n)) - P(G(n) \cap E(n)) = P(E(n-1))$$

$$\text{i.e. } p_n + q_n - P(\emptyset) = q_{n-1}$$

$$\therefore p_n + q_n = q_{n-1}$$

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MATHEMATICS Extension 2: Question.....

Suggested Solutions

Marks

Marker's Comments

Now, since $p_k = q_{k-1} - q_k$

for $k \geq 2$,

$$\sum_{k=2}^n p_k = \sum_{k=2}^n (q_{k-1} - q_k)$$

$$= (q_1 - q_2) + (q_2 - q_3) + \dots$$

$$\dots + (q_{n-2} - q_{n-1}) + (q_{n-1} - q_n)$$

$$= q_1 - q_n \quad (\text{telescoping series})$$

$$= 1 - q_n$$

Since $q_1 = P(\text{game does not end by 1st turn})$

$$= 1$$

as game must comprise at least two turns.

1

MATHEMATICS Extension 2: Question.....

Suggested Solutions

Marks

Marker's Comments

(iv) Consider $q_n = P(\text{game does not end by } n^{\text{th}} \text{ turn})$.

This is the case iff, upon conclusion of n^{th} turn, either

① all n turns are draws;

② A wins once, all other $(n-1)$ turns are draws;

③ B wins once, all other $(n-1)$ turns are draws;

④ A wins once, B wins once, all other $(n-2)$ turns are draws.

Scenario

Nb. ways can happen

① $\underbrace{\{D, D, \dots, D\}}_n : \frac{n!}{n!} = 1$

$$P(\underbrace{D, D, \dots, D}_n) = 1 \times \left(\frac{1}{6}\right)^n = \frac{1}{6^n}$$

② $\{A, \underbrace{D, D, \dots, D}_{n-1}\} : \frac{n!}{(n-1)!} = n \text{ ways,}$

so $P(\underbrace{A, D, \dots, D}_{n-1}) = \frac{n \cdot 5}{12} \cdot \frac{1}{6^{n-1}}$

MATHEMATICS Extension 2: Question.....

Suggested Solutions

Marks

Marker's Comments

$$(3) \text{ Similarly, } P(\underbrace{B, D, \dots, D}_{n-1}) = \frac{n \cdot 5}{12} \cdot \frac{1}{6^{n-1}}$$

$$(4) \{A, B, \underbrace{D, \dots, D}_{n-2}\} : \frac{n!}{(n-2)!} = n(n-1) \text{ ways,}$$

$$\text{so } P(\underbrace{A, B, D, \dots, D}_n) = n(n-1) \left(\frac{5}{12}\right)^2 \frac{1}{6^{n-2}}$$

Since all scenarios are mutually exclusive,

$$q_n = \frac{1}{6^n} + \frac{2 \cdot n \cdot 5}{12} \cdot \frac{1}{6^{n-1}} + n(n-1) \left(\frac{5}{12}\right)^2 \frac{1}{6^{n-2}}$$

$$= \frac{1}{6^n} \left[1 + \frac{6 \cdot 2 \cdot n \cdot 5}{12} + \frac{n(n-1) \cdot 25}{144} \cdot 6^2 \right]$$

$$= \frac{1}{6^n} \left[1 + 5n + \frac{25n(n-1)}{4} \right]$$

$$= \frac{1}{4 \times 6^n} \left[4 + 20n + 25n^2 - 25n \right]$$

$$= \frac{25n^2 - 5n + 4}{4 \times 6^n}$$

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MATHEMATICS Extension 2: Question.....

Suggested Solutions

Marks

Marker's Comments

Given $p_n = q_{n-1} - q_n$,

then

$$p_n = \frac{25(n-1)^2 - 5(n-1) + 4}{4 \times 6^{n-1}}$$

$$= \frac{-25n^2 - 5n + 4}{4 \times 6^n}$$

$$= \frac{1}{4 \times 6^n} \left[6(25n^2 - 5n + 25 - 5n + 5 + 4 - 25n^2 + 5n - 4) \right]$$

$$= \frac{1}{4 \times 6^n} (125n^2 - 325n + 200)$$

$$= \frac{25}{4 \times 6^n} (5n^2 - 13n + 8)$$

$$= \frac{25}{4 \times 6^n} (5n - 8)(n - 1) //$$

1